

Amphiphilic Balance: Effect of the Hydrophilic–Hydrophobic Ratio on Fouling-Release Surfaces

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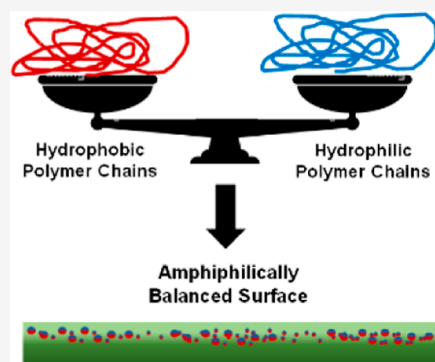


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ABSTRACT: This study demonstrated the importance of identifying the optimal balance of hydrophilic and hydrophobic moieties in amphiphilic coatings to achieve fouling-release (FR) performance that surpasses that of traditional hydrophobic marine coatings. While there have been many reports on fouling-release properties of amphiphilic surfaces, the offered understanding is often limited. Hence, this work is focused on further understanding of the amphiphilic surfaces. Poly(ethylene glycol) (PEG) and polydimethylsiloxane (PDMS) were used to create a series of noncross-linked amphiphilic additives that were then added to a hydrophobic-designed siloxane-polyurethane (SiPU) FR system. After being characterized by ATR-FTIR, XPS, contact angle analysis, and AFM, the FR performance was evaluated by using different marine organisms. The assessments showed that the closer the hydrophilic and hydrophobic moieties in a system reached a relatively equalized level, the more desirable the FR performance of the coating system became. A balanced ratio of hydrophilicity–hydrophobicity in the system at around 10–15 wt % of each component had the best FR performance and was comparable to or better than commercial FR coatings.



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INTRODUCTION

Marine biofouling is the accumulation of marine organisms on surfaces that are submerged in seawater, which is a complicated and challenging issue.^{1,2} There are more than 4000 different marine fouling organisms that attach in various ways and have different surface preferences.^{2–5} For instance, some organisms prefer hydrophilic surfaces, such as *Ulva linza* and mussels, while others are partial to hydrophobic surfaces or a combination of the two, such as *Navicula incerta* and *Cellulophaga lytica*. Furthermore, the buildup of biofouling is not linear, meaning that micro-organisms and macro-organisms have the potential to settle on surfaces in a dynamic pattern, further adding to the complexity of the issue.^{2,6–9} The economic and environmental impacts of marine biofouling on ships can be devastating. Its effects can include increased drag and reduced maneuverability, increased fuel consumption and emission of greenhouse gases, shortened service life, the transportation of nonindigenous and potentially invasive species to new aquatic environments, and increased maintenance costs.² For example, the US Navy spends an average of one billion dollars each year on the maintenance of their biofouling-affected ships.¹⁰ Dry-docking ships for routine cleaning to remove biofouling is not feasible, so alternative, reliable methods have been explored.

Biocide-containing antifouling (AF) paints have been widely used since the early 20th century, culminating in the highly effective tributyl tin-based and self-polishing coatings in the

1970s.² Unfortunately, these systems were recognized to be toxic to nontargeted marine organisms and were eventually banned worldwide.¹¹ Copper- and biocide-containing paints have been developed as replacements, but there has been a wave of investigations to find nontoxic alternatives. As an environmentally conscious replacement to AF paints, fouling-release (FR) technology has been explored as a method for maintaining a clean hull.^{12–16} These coatings rely on the weakened adhesion of marine organisms on a surface due to their inherent inhospitable nature, resulting in organisms being washed away with the movement of water vehicles. Traditional FR paints typically consist of hydrophobic surfaces that contain polydimethylsiloxane (PDMS) or fluoroalkyl polymers. However, these systems suffer from two issues that hamper their performance. First, these systems typically lack mechanical durability and require a tie-coat to facilitate their adhesion (which may not be very effective).^{2,6,17} Second, some biofouling organisms favor hydrophobic surfaces for settle-^{5,18}

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Table 1. List of Prepared Additives and Their Compositional Details

additive number	PDMS/PEG (wt % ratio)	PDMS (wt %)	PEG (wt %)	IPDI isocyanate (wt %)	PDMS (mol %)	PEG (mol %)	PDMS/PEG (molar ratio)
Amp-1	50:50	42	42	16	7.0	93.0	1:13
Amp-2	33:66	26	53	21	3.5	96.5	1:27
Amp-3	20:80	15	61	23	1.8	98.2	1:54
Amp-4	10:90	8	67	25	0.8	99.2	1:120
Amp-5	0:100	0.0	73	27	0.0	100.0	0:100

Table 2. Coating Compositions

formulation	added AmpAdd amphiphilicity				formulation amphiphilicity	
	additive type	additive amount (wt %)	PDMS (wt %)	PEG (wt %)	PDMS (wt %)	PEG (wt %)
A4		0.0	0.0	0.0	16.0	0.0
F1	Amp-1	20.0	8.4	8.4	24.4	8.4
F2	Amp-2	20.0	5.3	10.5	21.3	10.5
F3	Amp-3	20.0	3.0	12.6	19.0	12.6
F4	Amp-4	20.0	1.6	13.5	17.6	13.5
F5	Amp-5	20.0	0.0	14.6	16.0	14.6
T-10 control	none	0.0	0.0	0.0	10.0–12.0	10.0–12.0

To address the former, self-stratifying siloxane-polyurethane (SiPU) FR coatings were developed that have good adhesion and mechanical durability.^{17,19,26} The SiPU systems spontaneously separate into two layers by utilizing thermodynamics of incompatibility between polymer types, as well as overall surface energy minimization, providing a PDMS-rich surface and a PU-based bulk.²¹ However, to address the former, an introduction of hydrophilic domains to the interface is necessitated to deter the adhesion of hydrophobic-preferring organisms on a surface—known as amphiphilic surfaces containing both hydrophobic and hydrophilic domains.^{13–16,22–36}

While amphiphilic coatings have been investigated for nearly two decades, there is limited understanding of their mechanisms of performance, and design considerations have been very limited. In continuation of our previous work where we introduced the concept of critical amphiphilic concentrations (CACs) for amphiphilic surfaces,²³ the objective of this work was to further understand the necessary balance of hydrophilic and hydrophobic domains on a surface to attain optimal surface properties. To this end, we introduced a series of amphiphilic additives with increasing hydrophilic portion to a fully hydrophobic SiPU coating system, shifting the overall amphiphilic balance to assess our hypothesis.

The noncross-linked amphiphilic additives were composed of hydrophilic PEG and hydrophobic PDMS, attached to a polyisocyanate compound. The molecular weights of the PEG and PDMS were selected on the basis of published literature and our previous work.^{37,38} The weight ratios of the attached PEG and PDMS on the polyisocyanate were systematically varied to obtain additives of differing amphiphilicity, and the amphiphilic additives were added at 20 wt % to the base SiPU system. This specific amount of additive was selected based on the amount of PDMS present in the base SiPU formulation, which is 16 wt % (added as 20 wt % based on solid contents of acrylic polyol and IPDI trimer resin). Thus, the only variable explored in this study was the overall amphiphilic balance in the SiPU system, attained through the addition of the amphiphilic additives. Amphiphilic balance is the overall amount and ratio of hydrophilic and hydrophobic moieties in a formulation.

EXPERIMENTAL SECTION

Materials. Isophorone diisocyanate (IPDI)-based polyisocyanate Desmodur Z4470 BA was provided by Covestro LLC. Monocarbinol-terminated polydimethylsiloxane (PDMS) with molecular weight of 10,000 g/mol $\bar{M}_n$ (MCR-C22) was purchased from Gelest, Inc. Poly(ethylene glycol)methyl ether (750 g/mol $\bar{M}_n$), ethyl-3-ethoxy propionate, methyl ethyl ketone (MEK), acetylacetone, methyl amyl ketone (MAK), dibutyltin diacetate (DBTDAc), tetrahydrofuran (THF), and deuterated chloroform (CDCl_3) were purchased from Sigma-Aldrich. Toluene and 2-propanol were purchased from VWR. An acrylic polyol made of 80% butyl acrylate and 20% 2-hydroxyethyl acrylate was prepared via conventional free radical polymerization and diluted to 50% in toluene. Aminopropyl-terminated polydimethylsiloxane (APT-PDMS) with a molecular weight (MW) of 20,000 g/mol $\bar{M}_n$ was also synthesized through a ring-opening equilibration reaction. Both synthesized polymers were prepared following previously published guidelines from elsewhere.¹⁷

AkzoNobel International Paint provided the commercial FR standards Intersleek 700 (IS 700), Intersleek 900 (IS 900), and Intersleek 1100SR (IS 1100). Silicone elastomer Silastic T2 and polystyrene were provided by Dow Corning. Hydrophobic siloxane-polyurethane A4 coating (A4) as the base system for this study and was prepared following the procedure of Bodkhe et al. 2012.¹⁷ Also, a pure polyurethane formulation, without APT-PDMS, was prepared as a control using Desmodur Z4470 BA and the synthesized acrylic polyol. Amphiphilic T-10 coating, as an internal coating reference, was prepared following the procedure elsewhere for a formulation that contained 10 wt % PEG 750 g/mol $\bar{M}_n$ and PDMS 10,000 g/mol $\bar{M}_n$.³⁸ Aluminum panels (4 in. $\times$ 8" in., 0.6 mm thick, type A, alloy 3003 H14) purchased from Q-lab were sandblasted and primed with Intergard 264 (International Paints) using air-assisted spray application. Multiwell plates were modified using circular disks (1 in. diameter) of primed aluminum.

Design of Experiments. A series of amphiphilic additives (AmpAdd), based on PDMS and PEG, were synthesized and added to a hydrophobic SiPU coating system, called A4. The SiPU A4 system was composed of IPDI trimer isocyanate resin, cross-linked with acrylic polyol and amine-terminated PDMS, resulting in a self-stratified coating with PDMS on the surface and well-adhered PU on the bottom of an applied coating film. In A4, PDMS with 20,000 g/mol $\bar{M}_n$ was at 16 wt % based on total solids content of the formulation (added as 20 wt % based on solids contents of acrylic polyol and IPDI trimer resin). Thus, hydrophilic moieties that could diffuse into the surface need to be introduced to attain a coating with an amphiphilic surface.

AmpAdds were synthesized by reacting PEG and PDMS with the IPDI trimer polyisocyanate; the molar ratio of the isocyanate groups to the combined OH groups of PEG and PDMS was 1:1. These two were attached in varying weight ratios to meet the required one molar ratio while achieving several types of amphiphilicity. The molecular weights of 10,000 g/mol $\bar{M}_n$ for PDMS and 750 g/mol $\bar{M}_n$ for PEG were chosen to synthesize amphiphilic additives since desirable FR performance was reported and observed within these ranges.^{21,37,38} This study varied only the amount of attached PEG and PDMS on an additive, providing five modified additives that ranged from the least hydrophilic to most hydrophilic (Table 1). The table includes a contribution for IPDI isocyanate resin since the amphiphilic chains were grafted on its backbone. Thus, these wt % values of PEG and PDMS on an additive were used to determine the final content of PEG and PDMS moieties in a formulation.

Preliminary experiments indicated that systems having more than 20 wt % each of PDMS and PEG (combined 40 wt %) lacked mechanical integrity as a coating film, resulting in a crumbling coating film with poor adhesion to the substrate.³⁷ Knowing that SiPU A4 system contains up to 16 wt % PDMS and utilizing our prior experience on PEG-based additive systems,^{17,38,39} the AmpAdds of this study were added at only 20 wt % to achieve the desirable amphiphilic characteristics. As a result, a total of twelve formulations were investigated. Six formulations were A4 and its additively modified versions as outlined in (Table 2) and the other six were the internal and commercial references/standards (Table 3). Table 2

Table 3. List of Control Coatings

control name	control ID	description
amphiphilic SiPU	T-10	internal amphiphilic SiPU control
polyurethane	PU	pure polyurethane standard
polystyrene	PS	pure polystyrene standard (used for <i>U. linza</i> test)
Dow T2	T2	silicone elastomer standard
IS 700	700	Intersleek commercial FR standard
IS 900	900	Intersleek commercial FR standard
IS 1100	1100 SR	Intersleek commercial FR standard

outlines the type of added additive, amount of the added additive, wt % of PEG and PDMS based on the added additive, and overall wt % of PEG and PDMS in the total formulation (including the 16 wt % PDMS of A4). The balance of hydrophobic and hydrophilic contents in a formulation became more equal going from F1 to F5.

Commercial standards were prepared following the manufacturer's instructions. The T-10 internal amphiphilic control coating, containing 10,000 g/mol $\bar{M}_n$ PDMS and 750 g/mol $\bar{M}_n$ PEG prepolymer, was prepared following procedures given elsewhere; this formulation was selected since it showed the best performance among the studied systems.³⁸ As for the experimental coatings, all control and standards were prepared on 4" × 8" primed aluminum panels and multiwell plates. Table 3 contains descriptions of the control and standard coatings used for this study.

Synthesis of Amphiphilic Additive. The AmpAdd additives were synthesized by reacting hydroxyl-terminated PEG and PDMS chains with the IPDI trimer polyisocyanate (Scheme 1). The molar ratio of the NCO groups to the combined OH groups of PEG and PDMS was 1:1. Isocyanate groups were converted into urethane linkages by the attachment of PEG and PDMS chains. PEG and PDMS were added in weight ratios that met the required one molar ratio while accessing additives of varying hydrophilic–hydrophobic balance.

To synthesize Amp-3 (containing a weight ratio of PDMS/PEG 20:80), PEG 750 g/mol $\bar{M}_n$ (8.00 g) was diluted in toluene (8.00 g) in a 50 mL flask. The flask was equipped with a magnetic stirrer, nitrogen inlet, and temperature controller. A reflux condenser was used when heat was applied. PDMS 10,000 g/mol $\bar{M}_n$ (2.00 g) was added to the flask and vortex mixed for 2 min. IPDI trimer resin (4.25 g) and DBTDAC catalyst solution (1 wt % in MAK) (0.71 g) were then added to the flask. The reaction was carried out at 80 °C for 2 h. In theory, the synthesized prepolymer contained 61.0 wt % PEG and 15.0 wt % PDMS (Table 1). All other AmpAdds were synthesized by following the same procedure.

Isocyanate Titrations. Isocyanate titration was used to confirm the disappearance of isocyanate groups after the synthesis of the additive. An additive sample (0.3–0.5 g) was weighed in an Erlenmeyer flask and diluted with isopropanol. Then, 25 mL of 0.1 N dibutyl amine solution and 25 mL of isopropanol were added to the flask, and the mixture was stirred for 15 min. Several drops (3–5 drops) of bromophenol blue indicator were added and the content was titrated using a standardized 0.1 N hydrochloric acid until the end point blue to yellow was observed. A blank prepared only with 25 mL of a dibutylamine solution was also titrated following the same procedure. The recorded amount of hydrochloric acid for both titrations was used to calculate the isocyanate content (if any).

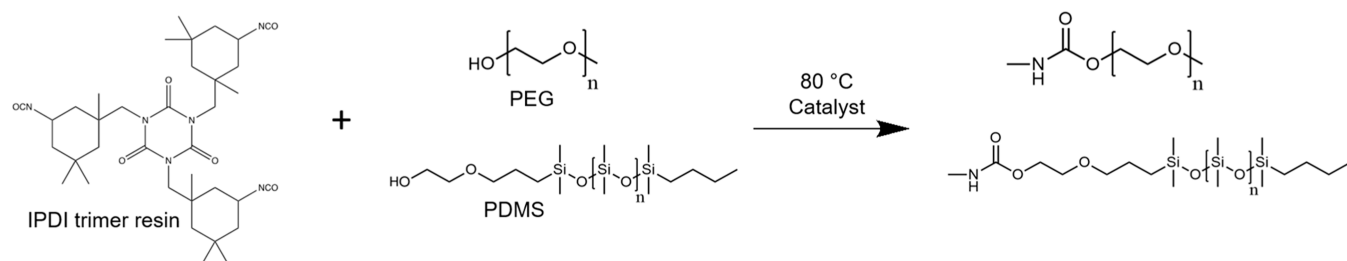
Percent Solids Determination. The nonvolatile content of the additive was determined following ASTM 2369. A weighed empty aluminum pan was filled with the additive sample (1–2 g). Isopropyl alcohol was used to cover the sample. The pan was placed in an oven at 120 °C for 1 h. After removal from the oven, the pan was weighed again to determine the percent solid. Three replicates were recorded.

Fourier Transform Infrared Spectroscopy. Fourier transform infrared (FTIR) spectroscopy (Thermo Scientific Nicolet 8700 FTIR) was used to characterize the additives. The additive was applied as a thin layer on a potassium bromide (KBr) plate to collect the spectrum.

Monte Carlo Simulation. DryAdd v5.0 software from OxMAT was used to estimate the compositions of the AmpAdd additives and determine how the PDMS and PEG chains were distributed on the base polyisocyanate forming the nonreactive additives.

Gel Permeation Chromatography. Gel permeation chromatography (GPC) analysis of the additives was run on an EcoSEC HLC-8320 GPC instrument from Tosoh Bioscience with a differential refractometer detector. Ten milligrams of samples was thoroughly dissolved in tetrahydrofuran (THF) in a vial. This was then filtered through a 0.45 mm PTFE filter before assessment. The samples were eluted at a flow rate of 0.35 mL/min. An EasiVial polystyrene standard obtained from Agilent was utilized for calibration.

Scheme 1. Overall Synthesis Scheme of Amphiphilic Additives



Proton Nuclear Magnetic Resonance Spectroscopy. ^1H NMR spectra were recorded in deuterated chloroform (CDCl_3) on a Bruker 400 MHz AvanceCore spectrometer. Topspin NMR analysis software from Bruker was utilized to process the data.

Coating Formulation and Curing. Coatings were prepared by adding the synthesized AmpAdds to the base SiPU A4 system. The A4 system is a hydrophobic marine fouling-release coating that is mainly composed of acrylic polyol, IPDI trimer isocyanate resin, and amine-terminated PDMS. The type of additive is the only variable among the formulations; the additives differed based on their degree of hydrophilic and hydrophobic balance.

For example, to formulate coating F3, acrylic polyol (15.06 g; 50% solid), acetyletacetone (1.39 g) (pot life extender), amine-terminated PDMS (2.32 g), and Amp-3 additive (5.93 g; 50% solid) were added to a vial and stirred at ambient conditions for 24 h. IPDI isocyanate trimer Desmodur Z4470 BA resin (5.55 g) and DBTDA catalyst solution (0.56 g) were added to the vial, and the mixture was stirred for another hour. Coating formulations were applied to primed $8' \times 4'$ aluminum panels using a wire-round drawdown bar with a film thickness of $80\ \mu\text{m}$. All coatings were cured under ambient conditions for 24 h, followed by oven curing at $80\ ^\circ\text{C}$ for 45 min. All experimental coatings were prepared by following the same procedure. Coatings were cut out in circular shapes and glued to the bottom of the wells of 24-well plates for biological assays.

X-ray photoelectron spectroscopy (XPS) was used to verify self-stratification of AmpAdd additives into the surfaces of coatings. To achieve this, two internal model PU systems, one without APT-PDMS (pure model PU) and one with Amp-1 additive in its composition (modified model PU), were also prepared by following the same procedure, except that the APT-PDMS cross-linker was omitted from both systems. These controls did not allow PDMS signals of A4 to interfere with PDMS signals of the additives for validation purposes.

Surface Characterization. A Kruss DSA 100 (drop shape analyzer) was utilized to measure the surface wettability and surface energy of the coatings. Three replicated measurements of water and diiodomethane contact angles were made for each sample. For each replicate, the static contact angle was measured over 9 min to monitor changes (the values plateaued after 9 min). Surface energy for each surface was calculated using the Owens–Wendt method.⁴⁰ Slip angle, advancing, and receding water contact angles of surfaces were evaluated using a tilting stage where a $25\ \mu\text{L}$ water droplet was viewed on a coating surface (tilted at $10^\circ/\text{min}$). The measured angles and surface energies were calculated using Kruss Advance software.

Attenuated total reflectance FTIR (ATR-FTIR) spectroscopy was used to characterize the surfaces of the coatings. A Bruker Vertex 70 with Harrick's ATR accessory with a hemispherical Ge crystal was utilized to collect ATR-FTIR spectra.

The elemental composition of coatings was determined using a Thermo Scientific K-Alpha X-ray photoelectron spectrometer (XPS). The instrument was equipped with monochromatic Al K_α (1486.68 eV) X-ray source and an Ar^+ ion source (up to 4000 eV). Depth profiling of a coating was evaluated by using Ar^+ etching with 300 eV cycles. For each etch cycle, the ion beam was set to 1000 eV monatomic mode with low current and 30 s etch time. After each etching cycle, five replicate survey spectra were collected, at low resolution, with a constant analyzer pass energy of 200 eV, for a total of 20 ms. For each run, photoemission lines for C 1s, N 1s, O 1s, and Si 2p were observed. Spectra were recorded at an angle normal to the surface (90°) of a $400\ \mu\text{m}$ area. The chamber pressure was maintained below 1.5×10^{-7} Torr, and samples were analyzed at ambient temperature. Atomic concentrations were quantified by the instrument's software as a representation of the atomic intensities as a percentage of the total intensity of all elements. Two internal PU systems, without (pure PU) and with Amp-1 additive (modified PU), were examined to verify self-stratification of AmpAdd additives into the surface of coatings.

Atomic force microscopy (AFM) was utilized to examine the surface topography of the studied coatings. A Dimension 3100 microscope with a Nanoscope controller scanned the surface of experimental coatings, collecting images on a sample area of $100\ \mu\text{m}$

$\times 100\ \mu\text{m}$ in the tapping mode. Scans were done in air under ambient conditions using a silicon probe with a spring constant of $0.1\text{--}0.6\ \text{N/m}$ and a resonance frequency of $15\text{--}39\ \text{kHz}$. For each surface, three replicate scans at different locations were collected to ensure the consistency and accuracy of the data.

Water Aging. All the prepared coatings were preleached for 28 days in running tap water. The water tanks were equipped to automatically fill and empty every 4 h. Water aging of the coatings is carried out to leach out any impurities that may interfere with fouling-release assessments and to determine the stability of coatings and surface rearrangements. All biological laboratory assays were carried out after the preleaching water aging process was completed.

Biological Laboratory Assays. All of the biological evaluations were conducted after 28 days of water aging. The coatings were assessed for leachate toxicity using *C. lytica*, *N. incerta*, and *U. linza* as discussed elsewhere.^{41,42}

The validated nontoxic coating systems were assessed against four model marine organisms including zoospores of macroalgae *U. linza*, bacterial cells *C. lytica*, diatoms *N. incerta*, and adult barnacle (*Amphibalanus amphitrite*). The details of experimental setups for these biological assessments can be found elsewhere.⁴³ The reported data for *U. linza*, *C. lytica*, and *N. incerta* is the biomass growth in and percent removal of the marine organisms' films. Alternatively, the reported data for *A. amphitrite* barnacles include the number of reattached barnacles on a surface and the average removal forces of such barnacles (note: due to the shortage of barnacles, we selected three coating systems to cover the range of assessed amphiphilic balanced systems and one commercial control which is known for its superior performance against barnacles.)

Statistical Analysis. Analysis of variance for this study was performed in SAS software, version 9.4. The GLM procedure with Tukey's and Dunnett's methods were utilized to determine the difference mean for each treatment group under the CRD (completely randomized design).

RESULTS AND DISCUSSION

The desire to eliminate the use of biocides in marine coatings has resulted in a shift toward hydrophobic fouling-release (FR) coatings. However, finding effective, durable, and nontoxic coating systems has proven to be a complex challenge, as the large number of fouling species means that there are usually some types that can take advantage of any new coating system. While hydrophobic FR coatings, such as silicone elastomers, or SiPU A4, offer relatively desirable performance, they lack the ability to prevent settlement of those biofoulants that prefer to settle and adhere to hydrophobic surfaces. Amphiphilic additives, containing hydrophilic and hydrophobic segments, are sought as a solution to improve the performance of hydrophobic FR coatings against a broader range of organisms. Thus, we prepared and investigated a series of amphiphilic additives (AmpAdd) in this study to enhance the performance of the hydrophobic A4 SiPU system and to correlate and understand the effect of the amphiphilicity balance on the properties of the coatings. The AmpAdds in this study were synthesized using a polyisocyanate that was modified with chains of $10,000\ \text{g/mol}$ $\bar{M}_n$ PDMS and $750\ \text{g/mol}$ $\bar{M}_n$ PEG. These molecular weights of PEG and PDMS were chosen due to their optimal FR performance, following reported literature and our previous studies.^{37,38}

The AmpAdds were synthesized with a relatively high MW PDMS ($10,000\ \text{g/mol}$ $\bar{M}_n$) and a low MW PEG ($750\ \text{g/mol}$ $\bar{M}_n$), where the weight ratio between PDMS/PEG ranged from 50:50 to 0:100. As the reagents do not react selectively with the IPDI-based polyisocyanate, this means that four possible types of AmpAdd triarmed molecules were created: IPDI-3PEG, IPDI-1PDMS-2PEG, IPDI-2PDMS-1PEG, IPDI-

3PDMS. Based on the very small molar ratio between PDMS and PEG (Table 1), the order of the AmpAdd triarmed molecule population is most probably IPDI-3PEG > IPDI-2PEG-1PDMS > IPDI-1PEG-2PDMS > IPDI-3PDMS. This can be confirmed with Monte Carlo simulations and experimental GPC measurements. The summarized data comparing the theoretical and experimental data are shown in Table 4.

Table 4. Monte Carlo Simulation vs GPC Experimental Data

(PDMS/PEG)	PDMS-dominated chains (30k MW & 21k MW)		PEG-dominated chains (12k MW & 3k MW)	
	theoretical wt % (Monte Carlo)	experimental wt % (GPC)	theoretical wt % (Monte Carlo)	experimental wt % (GPC)
Amp 1 (50:50)	6.2	4.0	93.8	96.0
Amp 2 (33:66)	2.1	2.9	97.9	97.1
Amp 3 (20:80)	0.6	1.6	99.4	98.4
Amp 4 (10:90)	0.2	1.2	99.8	98.8
Amp 5 (0:100)	0.0	0.0	100.0	100.0

The reaction that takes place to make the amphiphilic additives has varying weight ratios of PEG and PDMS and these molecules have quite different molecular weights. Therefore, the molar ratios are significantly different, and Monte Carlo simulations were run to determine the molecular weight distribution of the attached molecules on the IPDI polyisocyanate molecule. The likelihood of finding a molecule at a certain molecular weight and on the composition of the molecule is the output of the simulation. It was determined that even at a 50:50 weight ratio of PDMS and PEG almost half of the IPDI polyisocyanate molecules reacted with three PEG chains, while only over 6% is expected to have two or more PDMS molecules attached to the polyisocyanate. As the weight ratios are changed and a higher PEG concentration is incorporated, the fraction of molecules that have the majority PEG increases and the amount of PDMS majority molecules drops off significantly to almost zero. These results are expected as having a higher weight ratio of a lower molecular weight molecule means there are more moles to react with the IPDI polyisocyanate. The complete Monte Carlo simulation output is available in Table S1.

GPC was utilized to compare the theoretical Monte Carlo simulation results with the experimental results. While we expected to observe four MW distributions in the GPC output, the results only showed two MW ranges. However, when looking at the start and end molecular weight of these two distribution curves, it was determined that the first peak to elute contains the higher molecular weight portions of PDMS-dominated chains (30k and 21k from the simulation data). The second peak shows that it contains the lower-molecular-weight portions of PEG-dominated chains (12k and 3k from simulation data). The area under the curves is also shown and gives a general idea of how many molecules are present at that given distribution curve. These areas match very closely with the data of the Monte Carlo stimulation, as shown in Table 4. As the PEG concentration is increased, the area and

size of the first peak increase, while the size of the second peak decreases. This correlates with the simulation data and shows that a higher PEG concentration causes more PEG-dominated polyisocyanate molecules. The GPC experimental data are provided in Table S2 with the GPC curves in Figure S1.

The AmpAdds were prepared by reacting OH-terminated PEG and PDMS with an IPDI-based polyisocyanate. The complete disappearance of the isocyanate groups, as determined by FTIR and isocyanate titration, confirmed the formation of the urethane linkage and the attachment of the amphiphilic compounds. FTIR spectra of all five AmpAdds show the disappearance of the isocyanate signal at 2250 cm^{-1} and a broadened stretching for secondary amine (from the formed urethane linkage) at 3350 cm^{-1} (Figure 1), confirming

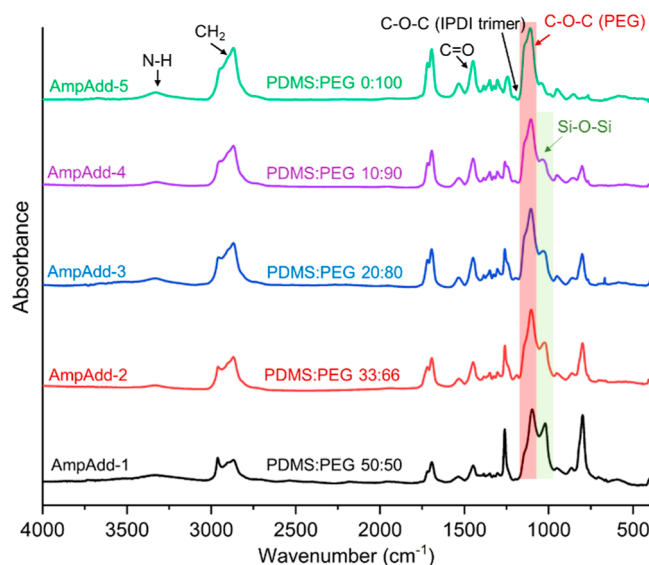


Figure 1. FTIR spectra of the five amphiphilic additives. Each spectrum is labeled with the additive ID and its PEG and PDMS compositional ratio.

the complete transformation of the available isocyanate groups. Furthermore, the presence of the Si–O–Si signal at 1035 cm^{-1} (Figure 1; highlighted red) and C–O–C at 1105 cm^{-1} (Figure 1; highlighted green) confirm the attachment of PEG and PDMS chains on the additives. The intensity ratio of Si–O–Si and C–O–C peaks changed with the composition of the AmpAdd. As more PEG was attached on an additive, the ether peak increased, while the siloxane peak became less visible until it completely vanished in Amp-5 (which contained only PEG chains). Additionally, isocyanate titrations on the synthesized additives further showed that no isocyanate groups remained. The synthesis was carried out using dried toluene to prevent the reaction of isocyanates with water.

Proton NMR was utilized to further characterize the amphiphilic additives and see how the changing concentration affected the spectra. As with the FTIR peaks changing in intensity as the concentration changes, so do the repeat units for PEG and PDMS. By setting the peak integration to 1 for the peaks occurring at 1 ppm, the repeating units for PDMS (0.1 ppm) and PEG (3.7 ppm) can be compared (Table 5). As the PEG concentration increases, the peak at 3.7 ppm increases in intensity and the peak at 0.1 ppm decreases in intensity (Figures S2–S6). The calculated ratios of signature NMR peaks of PEG and PDMS chains correlate closely with the

Table 5. Integration Values of ^1H Content for Repeat PDMS (0.1 ppm) and PEG (3.7 ppm) Repeat Units

(PDMS/PEG)	PDMS (0.1 ppm)	PEG (3.7 ppm)	NMR calculated PDMS/PEG ratio
Amp 1 (50:50)	3.33	3.77	47:53
Amp 2 (33:66)	1.74	4.07	30:70
Amp 3 (20:80)	0.98	4.44	18:82
Amp 4 (10:90)	0.46	4.59	9:91
Amp 5 (0:100)	0	4.4	0:100

theoretical ratios used originally in the synthesis, confirming that the AmpAdd additives were synthesized as expected/ designed (Table 5).

Coatings were prepared containing the amphiphilic additives, and the surfaces of the coatings were characterized with ATR-FTIR, contact angle measurements, and AFM, while the self-stratification of additives was confirmed with XPS using control coatings. ATR-FTIR of Amp-modified A4 systems showed the signal for C–O–C of PEG at 1100 cm^{-1} that was absent for the unmodified A4 coating (Figure 2;

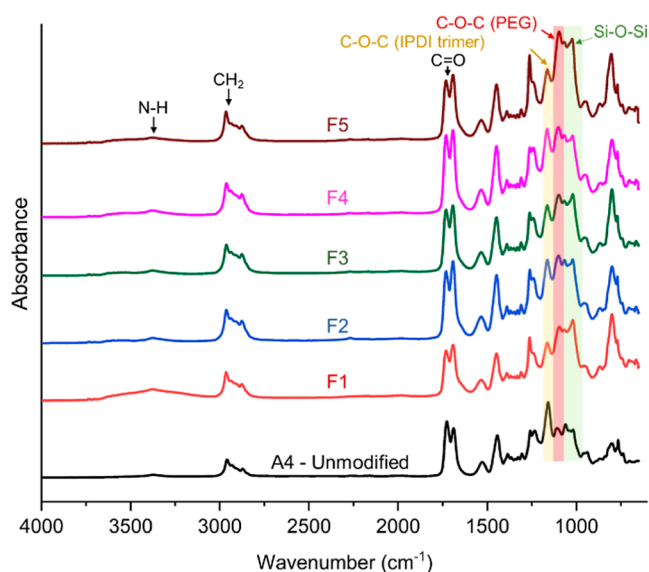


Figure 2. ATR-FTIR spectra of the unmodified and modified A4 coatings. Green, red, and yellow highlights reflect the peaks of interest for siloxane, ether from PEG, and ether from urethane linkage, respectively. Each spectrum is labeled to reflect the formulation type.

red highlight). The intensity of the Si–O–Si peak at 1016 cm^{-1} also increased with the addition of AmpAdds. The relative intensities of ether and siloxane peaks correlated with the types of utilized AmpAdd. For example, F1 (having Amp-1) displayed a strong siloxane signal, while F5 (having Amp-5) exhibited the strongest ether (of PEG) signals among all coatings. The remaining signature peaks were present among all systems including ether of urethane (from additives and cross-linked coating) at 1160 cm^{-1} , carboxyl at $1688\text{--}1727\text{ cm}^{-1}$, and urethane NH at 3365 cm^{-1} .

Static and dynamic contact angle measurements were carried out to evaluate the interaction of water and diiodomethane with the prepared FR surfaces. The static contact angle

measurements were carried out over 9 min (after this the collected values plateaued). The A4 system did not present any significant changes in water contact angles (WCAs) and diiodomethane contact angles (MICAs) over time, whereas on Amp-modified A4 coatings WCAs and MICAs decreased (Figure 3A). This dynamic change of contact angles as a function of time is similar to that of control amphiphilic T-10, suggesting that the hydrophilic domains on the surface of modified SiPU A4 (from added amphiphilic additives) swell upon exposure to water and allow the water droplet to spread more on the surface. The comparison of formulations indicates that WCAs drop over time as more hydrophilicity is introduced into the system and also that the change in values is more noticeable for WCAs than MICAs. Surface energies for the coatings were calculated using WCAs and MICAs (Figure 3B). The additives caused a considerable increase in surface energies for the modified coatings in comparison to those of unmodified A4. The surface energy (SE) for the Amp-modified coatings initially ranged around $40\text{--}50\text{ mN/m}$ and increased by $5\text{--}10\text{ mN/m}$ over 9 min (except F5 coating), while original A4 displayed a relatively stable SE around $27\text{--}30\text{ mN/m}$ (which is a typical value for PDMS rich surfaces). Unlike the other modified formulations, F5 showed contact angles and surface energies similar to those of the amphiphilic T-10 control. This different behavior is attributed to the similarity in composition of the F5 and T-10, since both contain equal contents of PEG and PDMS. While it is reported that fouling-release coatings with surface energy of about $22\text{--}24\text{ mN/m}$ can reduce interactions between the marine organism and the substrate,⁴⁴ our work in this study and previous studies appears to show that this theory does not necessarily hold true for amphiphilic surfaces.^{25,38,39,45}

The dynamic WCA experiments using a tilting stage showed that the slip angle (water droplet roll-off angle) of the original A4 significantly decreased upon the introduction of the amphiphilic additives, from 9 to $2\text{--}3^\circ$ (Figure 3C). Once the PEG and PDMS balance was equalized in the F5 system, its slip angle was comparable to the amphiphilic T-10 control.^{38,46–48} Overall, the decreased roll-off data indicate that the amphiphilic additives improve the FR properties of the A4 system to remove objects from its surface. Advancing contact angles (Adv CA) and receding contact angles (Rec CA) for the modified A4s (except F5) were in a much lower range (around $40\text{--}60^\circ$) than the unmodified A4 coating ($95\text{--}105^\circ$), while the F5 system again displayed a behavior similar to T-10 (Figure 3D). The low contact angles of the modified coatings implied the presence of hydrophilic domains on the surfaces. Additionally, it is desirable to have a CA hysteresis (the difference between Adv CA and Rec CA) of less than 10° since this is the indication of a smooth surface; the assessed coatings demonstrated this behavior as well. After-immersion coating-specific trends in contact angles were observed (28 days water aging) (Figure S7). For example, coatings A4 and F5 showed a decrease for WCAs, and the remaining coatings showed an increase for WCAs. The changes may be related to several factors including water absorption, rearrangements of nonbound amphiphilic domains, and possible leaching of the additives.

AFM was employed to investigate the effects of the additives on the surface morphology of the A4 system (Figure 4). Generally, soft materials like PDMS appear lighter (high phase angles), and harder materials like PEG and PU matrix moieties will appear darker (low phase angles). The A4 system

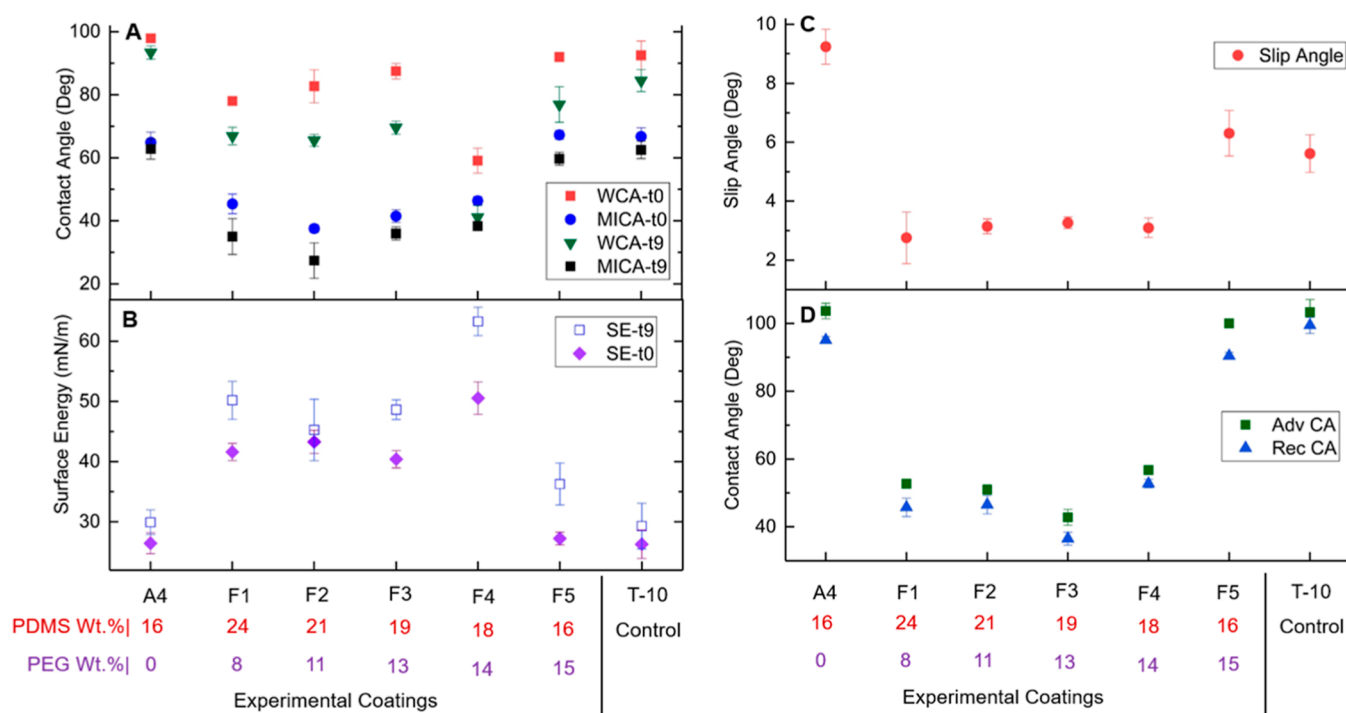


Figure 3. Contact angle and surface energy data for coatings: (A) water contact angles (WCAs) and methylene iodide contact angles (MICAs) at 0 and 9 min; (B) surface energy (SE) of coatings at 0 and 9 min, calculated by the Owens–Wendt method utilizing the average WCAs and MICAs for each coating; (C) slip angle of coatings where a water droplet starts to roll off; and (D) advancing contact angle (Adv CA) and receding contact angle (Rec CA) data, measured by tilting method. T-10 is the internal coating for comparison. The X-axis is labeled to reflect overall content of PEG and PDMS in a formulation. The error bars are the standard deviation of three measurements.

displayed a homogeneous surface with no apparent domains, which is due to its surface composition being solely PDMS. The introduction of the additives caused the appearance of heterogeneous domains on the surface of the coatings. The surface heterogeneity is a result of thermodynamically driven self-segregation and incompatibility between PDMS and PEG moieties. This behavior was expected as it was previously reported for other amphiphilic systems such as the T-10 control system, confirming the presence of amphiphilic domains on the surfaces for this study.³⁸ The morphologies differed among the studied coatings, which can be attributed to the type of incorporated additive. As the amount of hydrophilicity in an additive increased, the surface domains appeared to be smaller and present at a higher distribution throughout the surface. This trend can be observed from coatings F1 to F4. High quality images for F5 could not be captured, which was attributed to the nature of the surface after introduction of the additive. Furthermore, AFM images were recorded for the coatings after 28 days of water immersion. As expected from the postimmersion contact angle data, the AFM images supported there was no remarkable decrease of the domains from the surfaces due to potential leaching of the additives (Figure S8). The AFM images supported the ATR-FTIR data and contact angle measurements in which AmpAdds were present on the surface and contribute to the dynamic interaction of the surfaces with the assessed droplets. An assessment of surface roughness of the coatings suggested the systems became less rough as the amphiphilicity balance was attained which may have been one of the key parameters in observing desired FR performance as discussed in analyzing the biological organisms in the latter sections (Table S3).

XPS experiments were used to validate the self-stratification of the amphiphilic additives on the surfaces of coatings. Our previous work showed that the self-stratified nature of the amphiphilic SiPU and A4 system can be observed via XPS.³⁹ Initially, the XPS measurements were carried out on the modified and unmodified A4 coating systems as a function of depth; however, it was impossible to distinguish between Si atoms arising from the additives and the A4 formulation itself. In order to isolate and understand the stratification of the amphiphilic additives, a model PU system was utilized to validate diffusion of the additives into the surface. The model PU system was the A4 formulation without amine-terminated PDMS, and for validation, the Amp-1 additive was added to the model PU formulation at 20 wt %. The depth profiling XPS analysis of the model formulations (PU and the Amp-modified PU) confirms that the amphiphilic additive self-stratified to the surface (Figure 5). For the model PU, the concentration of carbon, oxygen, and nitrogen atoms was almost constant throughout the coating, signaling a homogeneous network in terms of elemental composition (Figure 5A). Alternatively, the model modified PU system contained Si atoms (due to the addition of Amp-1) on its surface, which gradually decreased as a function of depth, while elemental compositions of carbon and nitrogen increased (related to the bulk of the coating). The estimated atom percentages for additive Amp-1 are as follows: C: 53.9%; O: 24.4%; Si: 20.7%; and N: 0.9%. These values are very close to what is observed at 0 etch depth. Therefore, the XPS data indicated that self-stratification of the additives occurred, resulting in amphiphilic surfaces, consistent with data from ATR-FTIR, contact angle assessments, and AFM.

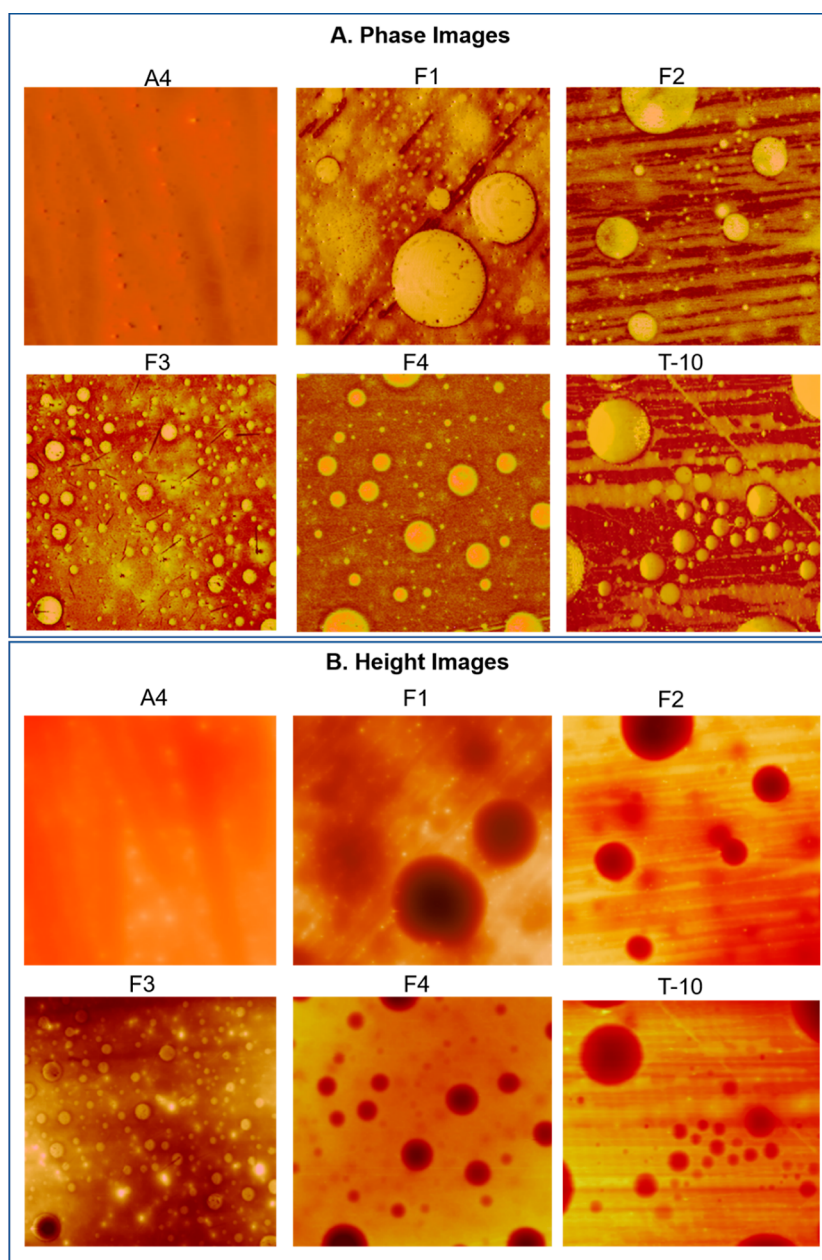


Figure 4. AFM phase images (top box) and height images (bottom box) of unmodified and modified SiPU A4 and T-10 coatings. Each image is for an area of $100\ \mu\text{m} \times 100\ \mu\text{m}$. Each label reflects the coating number.

To evaluate the effect of the AmpAdds on the fouling-release performance of the coatings, they were evaluated using a series of representative marine organisms. All of the evaluations were conducted after 28 days of water aging. The leachates from all coatings were nontoxic, allowing for valid FR experiments.

The biomass of *U. linza* developed over 7 days was relatively similar on all experimental coatings, internal controls, and commercial systems (Figure S9; comparison *P*-values >0.05 , Tukey's method). The fact that the initial biomass of *U. linza* was not altered by the addition of the AmpAdds indicated that the amphiphilic/hydrophilic additives did not result in a surface that would deter initial settlement of the algae, unlike the reported literature.⁴⁹ The biofouled surfaces were water-jetted at water pressures of 10 and 16 psi, and the percent removal/release is shown in Figure 6 (10 psi blue bars, 16 psi green bars). The release trend was similar among coatings at

both pressure levels but, not surprisingly, the 16-psi pressure level released more *U. linza* from the surfaces (comparison *P*-values <0.05 , Tukey's method). The introduction of AmpAdd additives to A4 considerably improved its FR performance at both 10 and 16 psi (comparing A4 with F1–F5; comparison *P*-values <0.05 , Tukey's method). The extent of release for Amp-modified coatings at 10 psi was similar, while the extent of release varied among coatings at 16 psi. Coating F3 outperformed all the studied coatings and offered almost 60% improvement over the base A4 system (comparison *P*-values <0.05 , Tukey and Dunnett methods). Comparing the performance of coatings with internal and commercial controls, it was evident that the addition of AmpAdd improved the performance of the base A4 system, boosting the *U. linza* release to be comparable to or better than the references. Coatings F1, F3, F4, and F5 outperformed the well-performing

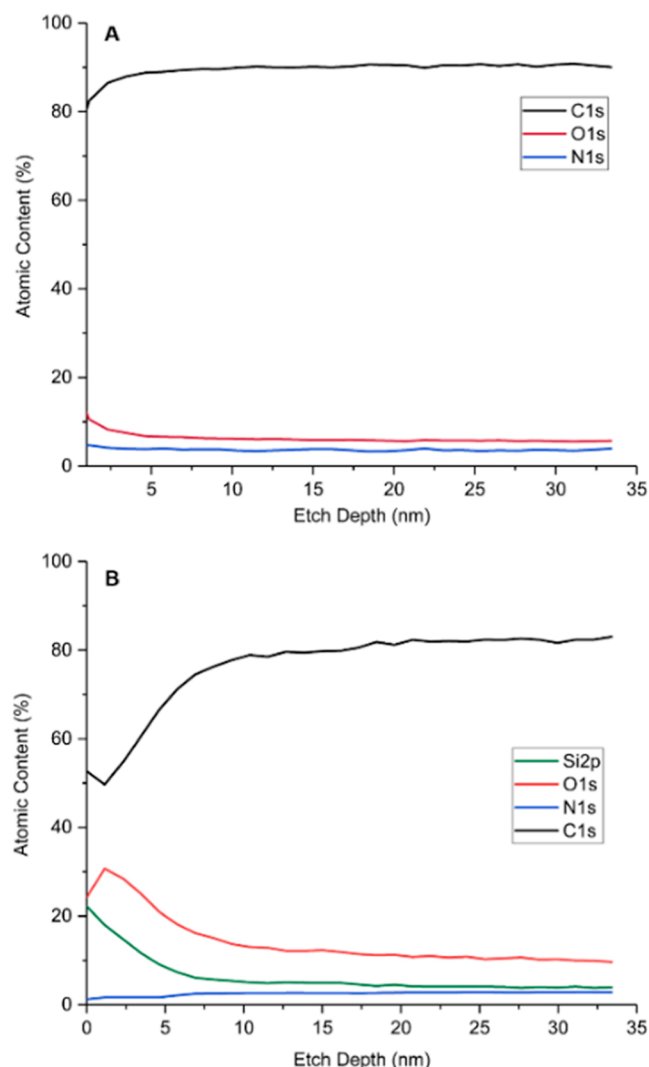


Figure 5. XPS data: (A) XPS depth profiling data for unmodified PU coating; and (B) XPS depth profiling data for PU coating containing 20 wt % Amp-1.

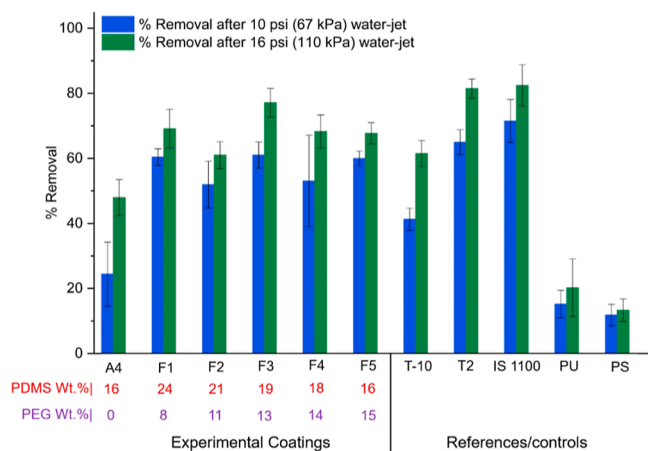


Figure 6. *U. linza*: percent removal/release at 10 psi water-jet (blue bar) and 16 psi water-jet (green bar). The X-axis is labeled to reflect overall content of PEG and PDMS in a formulation. Each category of assessed coatings is separated with lines. The error bar is the standard deviation of six measurements.

amphiphilic T-10 (comparison P -values <0.05 , Tukey's method), and coating F2 displayed a comparable performance (comparison P -values >0.05 , Tukey's method). Coating F3 also showed a matching performance to the commercial coatings, T2 and 1100 SR (comparison P -values >0.05 , Tukey's method). These results indicate that imparting amphiphilic character to the base A4 system resulted in significantly improved performance, which highlights that having an amphiphilic surface is beneficial to release a greater extent of *U. linza* from the surface. The observation is in consistent with reports that *U. linza* preferably settles at a higher rate on hydrophobic surfaces compared to hydrophilic surfaces.^{7,9,50} Overall, the FR data for *U. linza* imply that the addition of amphiphilic additives improves the performance of the hydrophobic SiPU A4 system.

Unlike the results for *U. linza*, *C. lytica* settled significantly less on the experimental systems in comparison to the commercial systems (comparison P -values <0.05 , Tukey's method), while they grew almost equally on both experimental and internal control/reference coatings (comparison P -values >0.05 , Tukey's method) (Figure S10). The settlement mass of *C. lytica* was not changed for additively modified coatings compared to the control base A4, indicating the additives did not have a detrimental effect (except for formulation F2). *C. lytica*'s unaltered biomass settlement after the addition of the AmpAdds indicated that the amphiphilic/hydrophilic additives did not result in a surface with further fouling-resistance properties.⁴⁹ The biofouled samples were water-jetted at two pressure levels, and the removal/release of *C. lytica* film was plotted for 10 psi (Figure 7—blue bars) and 20 psi (Figure

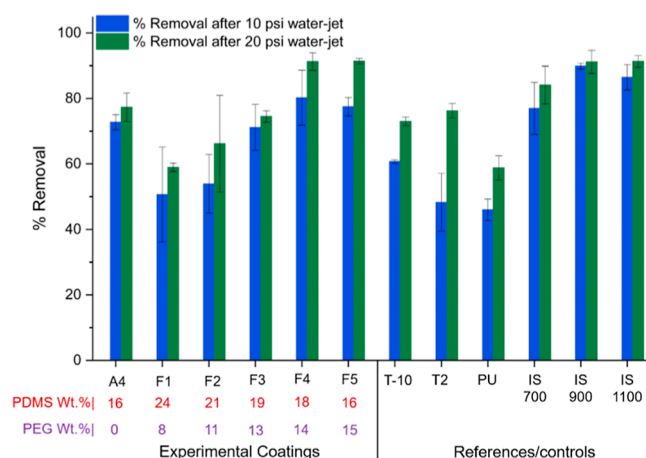


Figure 7. *C. lytica*: fouling-release data for percent removal/release at 10 psi water jet (blue bar) and at 20 psi water jet (green bar). The X-axis is labeled to reflect overall content of PEG and PDMS in a formulation. Each category of assessed coatings is separated with lines. The error bar is the standard deviation of six measurements.

7—green bars). The higher water pressure level released more *C. lytica*, but the overall trend among the coatings remained unchanged. The addition of Amp-1 and Amp-2 was detrimental to FR performance of A4 (formulations F1 and F2) (comparison P -values <0.05 , Tukey's method). This diminished performance was attributed to the increased hydrophobic content of the A4 systems by hydrophobic-inclining AmpAdds, while the overall hydrophilic content of the formulations remained relatively low (Table 2). However, the FR performance for Amp-modified systems improved

when additives that contained a greater hydrophilic content were added to the A4 system, resulting in *better* amphiphilically balanced systems (Table 2). Coatings F4 and F5 contained almost equal amounts of PEG and PDMS in their composition after adding the AmpAdd additives. Coatings F4 and F5 delivered remarkably better performance than A4 at 20 psi (comparison P -values <0.05, Tukey's method), while coating F3 offered a matching performance to A4 (comparison P -values >0.05, Tukey's method). The well-balanced amphiphilic systems outperformed the internal amphiphilic reference T-10 (comparison P -values <0.05, Tukey's method). Also, the performance of F4 and F5 systems matched the top-performing commercial coatings such as IS 900 and IS 1100 (comparison P -values >0.05, Tukey's method) and outperformed all the other assessed reference coatings (comparison P -values <0.05, Tukey's method). The *C. lytica* data clearly suggest that the more equally balanced the hydrophilic and hydrophobic moieties are in a system, the better the FR performance. These results demonstrate consistency with the reported literature where *C. lytica* is recognized as a microbiofoulant with an affinity to settle on a wide range of surfaces, ranging from hydrophilic to hydrophobic,² and amphiphilic surfaces are known to be a good deterrent against this organism.³⁸ Overall, the FR data with *C. lytica* suggest achieving an amphiphilically balanced system is key for desired FR performance based on F4 and F5 systems.

The extent of *N. incerta* biofouling was comparable among the experimental coatings and the internal controls, including the base A4 system (comparison P -values >0.05, Tukey's method) (Figure S11). The additively modified A4 systems showed considerably less fouling than the T2 and IS 700 commercial systems, but significantly higher fouling than commercial controls IS 900 and IS 1100 (comparison P -values <0.05 for both comparisons, Tukey's method). Similar to *C. lytica* and *U. linza*, the surfaces fouled by *N. incerta* were water-jetted at two pressure levels and removal/release of the *N. incerta* film was reported at 10 psi (Figure 8—blue bars) and 20 psi (Figure 8—green bars). The addition of AmpAdds for all experimental formulations (F1–F5 coatings) significantly improved the FR performance of the A4 system at both 10 and

20 psi (comparison P -values <0.05, Tukey's method). The FR performance of modified coatings appeared to be slightly better when a small amount of hydrophilicity was present in a system. Modified coating systems offered a matching performance to internal PU and IS 700 commercial coating (comparison P -values >0.05, Tukey's method) but were not as good as IS 900 and IS 1100 (comparison P -values <0.05, Tukey's method). The *N. incerta* data reveal that introducing amphiphilicity to the hydrophobic A4 system contributed to enhanced performance; however, there is no indication that a particular amphiphilic balance level was better than another (comparing F1–F5 coatings among each other, P -values >0.05, Tukey's method). The result is in agreement with reported literature that *N. incerta* is a recognized biofouling organism that generally adheres more strongly to hydrophobic surfaces than to hydrophilic ones.^{9,50} Therefore, it can be inferred that the introduction of the amphiphilic/hydrophilic additives to the base hydrophobic A4 system resulted in a higher release rate of the *N. incerta* from the surface.

The addition of the amphiphilic additives did not result in reduced initial adhesion of the *A. amphitrite* barnacles for all of the assessed systems. For example, the number of barnacles that initially adhered to F1, F3, F5, and A4 SiPU coating systems out of a sample of six barnacles was four, five, six, and five, respectively (note: due to a limited supply of barnacles, we selected three coating systems to cover the range of assessed amphiphilic balanced systems and one commercial control which is known for its superior performance against barnacles.)

The fouling-release data showed that when the fully hydrophilic Amp-5 additive was introduced to the A4 formulation, it resulted in a well-balanced amphiphilic surface that significantly reduced the adhesion strength of barnacles (Figure 9; comparison P -values <0.05 with A4 system, Tukey and Dunnett methods). Coating F5 also showed better performance (less force required to release reattached barnacles) than the commercial IS 1100, which is especially recognized for its good performance against barnacles (comparison P -values <0.05, Tukey's method). A direct correlation was observed between FR performance and amphiphilic balance. The data highlighted when the system is not well-balanced amphiphilically, its performance is not desirable such as F1 and F3; however, when a system possesses relatively equal amounts of hydrophilic and hydrophobic domains, its performance is not compromised against barnacles due to amphiphilicity (Figure 9). This enhanced performance for a balanced amphiphilic surface is in accordance with other published results.³⁸ The amphiphilic surfaces did not significantly reduce the number of reattached barnacles on a surface (comparison P -values >0.05, Tukey's method). Different species of barnacles have different affinities for surfaces; thus, it is challenging to propose an underlying rule for their preferred surface type.^{2,51–54} For example, *A. amphitrite* settles on hydrophilic surfaces as well as hydrophobic surfaces.^{55–58} While the barnacle data did not conclusively show that amphiphilicity was greatly beneficial against *A. amphitrite*, it did demonstrate that once an amphiphilic well-balanced surface is presented to the organism, its adhesion to the surface is comparable to a fully hydrophobic system and unbalanced amphiphilic surfaces can be detrimental, thus highlighting the importance of designing a surface with an optimal combination of hydrophilic and hydrophobic domains.

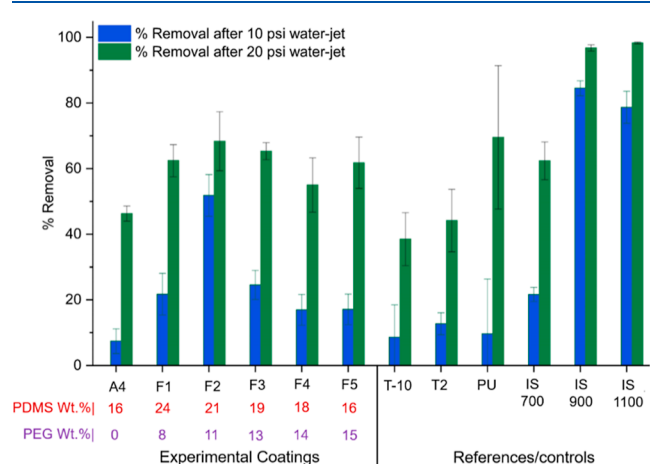


Figure 8. *N. incerta*: fouling-release data for percent removal/release at 10 psi water-jet (blue bar) and 20 psi water-jet (green bar). The X-axis is labeled to reflect overall content of PEG and PDMS in a formulation. Each category of assessed coatings is separated with lines. The error bar is the standard deviation of six measurements.

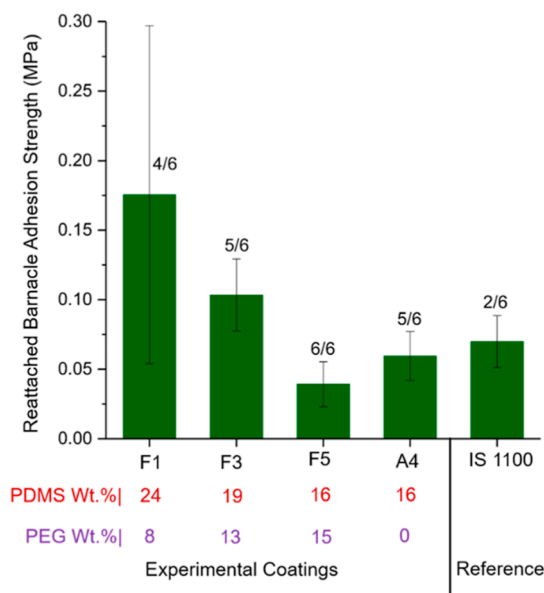


Figure 9. Reattached barnacle (*A. amphitrite*) adhesion strength. Six barnacles were used for each reattachment study. The number of reattached barnacles out of six is labeled for each system as a ratio. Each bar shows the average adhesion strength based on the number of successfully reattached barnacles. The X-axis is labeled to reflect overall content of PEG and PDMS in a formulation. Each category of assessed coatings is separated with lines. The error bar represents the standard deviation of the measurements that were assessed on reattached barnacles.

CONCLUSIONS

This study investigated the effect of amphiphilic balance on a fouling-release surface performance by the synthesis of a series of amphiphilic additives by installing PEG and PDMS chains on a polyisocyanate backbone via a facile and easily scalable procedure. The additives were prepared with differences in their degree of amphiphilicity, ranging from the least hydrophilic to most hydrophilic, confirmed by Monte Carlo simulations, GPC, and ^1H NMR assessments. This study demonstrated that a hydrophobic marine coating can be converted into a desirably performing amphiphilic coating by utilizing amphiphilic additives. The introduction of AmpAdds into the SiPU A4 system resulted in an amphiphilic surface, as supported by ATR-FTIR, AFM, XPS, and contact angle measurements. ATR-FTIR spectroscopy showed the presence of both PEG and PDMS signals. AFM detected the formation of heterogeneous microdomains that were missing on the surface of the original A4 coating. XPS analysis demonstrated that the additives self-stratify into the surface of a coating system, providing an amphiphilic surface, while contact angle measurements supported the presence of amphiphilicity through the dynamic interaction of modified surfaces with assessed droplets. Biological assays strongly supported that the AmpAdds improved FR performance of the SiPU A4 systems against macroalgal spores (*U. linza*), bacteria (*C. lytica*), and diatoms (*N. incerta*), while they were detrimental (F1 and F3 systems) or did not alter/slightly enhance the performance (F5 system) against barnacles (*A. amphitrite*). The FR performance of the modified A4 coating systems was in most cases comparable to top-performing commercial marine paints or even better for specific organisms. Overall, this study demonstrated coating systems that were well-balanced in

terms of hydrophilic and hydrophobic contents performed better than systems that were mainly hydrophobic.

The importance of the amphiphilic balance demonstrated in this study and our previously published concept of CAC³⁷ highlight the need for a fundamental understanding of amphiphilic surfaces to improve their effectiveness in coatings used in the marine industry, medical devices, and anti-icing surfaces. Our future efforts will examine the amphiphilicity balance concept for other applications and the effect of switching the hydrophilic group from introducing PEG to other hydrophilic materials (i.e., zwitterions).

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.langmuir.3c03478>.

Contact angle measurements, AFM images for before and after water aging immersions, details of Monte Carlo simulations, and initial settlement/biofouling data for marine organisms (PDF)

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Notes

The authors declare no competing financial interest.

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